



Start and Go

Quick facilitation notes & answer key

Goal

Students read a short block program and run it on a number path to find where Bit lands, then alter one block and predict the new result. This is the reading-and-running half of coding (C3.1) plus a first taste of altering code and describing the effect (C3.2).

Ontario curriculum (Grade 1 — C3 Coding)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential events.

C3.2 — read and alter existing code, including code that involves sequential events, and describe how changes to the code affect the outcomes.

Materials

The printed worksheet and a pencil. Optional: a floor number path 0-6 (tape and number cards) so a child can walk each program.

Run it (about 20 min)

1. I DO: run Code A aloud — 'start at 0... forward 2 lands on 2... forward 1 lands on 3.' One step per block, top to bottom.
2. WE DO: Exercise 1 — students run Code A to 3. Exercise 2 — predict Code B's landing (4), then run from 1 to confirm.
3. YOU DO: Exercise 3 — change forward 3 to forward 2; predict and run to land on 3.
4. Connect: 'Code runs in order from the top. The start block must go first so Bit knows where to begin.'

Answer key (modelled by the run-gate)

Ex 1 — Code A: 0 → (forward 2) 2 → (forward 1) 3. Bit lands on 3.

Ex 2 — Code B: 1 → (forward 3) 4. Bit lands on 4.

Ex 3 — change to forward 2: 1 → 3. Bit lands on 3.

Why the order works: the start block runs first to set Bit's number; if a move block ran before the start, Bit would have no number to move from and the code would break. The moves then run top to bottom, one at a time.

Watch for & differentiate

Common slip: counting the start number as the first step — 'start at 0' is a position, not a move; the first forward step leaves 0.

Simplify: walk each program on a floor number path, pausing on each number.

Extend: ask students to change the start block to 'start at 2' and predict Code A's new landing (5).

Success indicator: the child runs the blocks in order and writes the correct landing number for each code.